

Atlas Product Requirements Document (PRD)

Version 0.1 – Foundation

Product Name: Atlas

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1. Executive Summary

Atlas is an AI-powered engineering learning ecosystem designed to help learners become professional software engineers through guided practice, real-world software engineering missions, AI mentorship, and production-inspired development workflows.

Atlas focuses on engineering thinking rather than only algorithmic problem solving.

2. Product Goals

Primary Goals

- Make beginners job-ready.

- Replace fragmented learning.
- Build engineering thinking.
- Create portfolio-worthy projects.
- Improve developer confidence.

Secondary Goals

- Corporate training.
 - University partnerships.
 - Hiring platform.
 - Certification ecosystem.
-

3. Product Principles

Every feature added to Atlas must satisfy at least one of these principles.

- Learn by Building
 - Engineering First
 - Think Before Code
 - Guide Instead of Solving
 - AI as Mentor
 - Practical over Theoretical
 - Portfolio Driven
 - Industry Ready
-

4. User Roles

Student

Learns, solves missions, earns XP, builds portfolio.

Mentor

Reviews work, creates learning content, provides feedback.

Company

Publishes assessments, recruits talent, reviews portfolios.

Administrator

Manages users, missions, AI prompts, analytics, moderation.

5. User Journey

A learner follows this progression:

Register



Select Goal



Choose Learning Path



Complete Lessons



Solve Logic Missions



Build Features



Debug Applications



Receive AI Feedback



Build Portfolio

↓

Earn Certification

↓

Interview Ready

↓

Employment

6. Module Specifications

Every module in Atlas will follow this specification template.

Module Name

Purpose

Business Goal

User Story

User Flow

Functional Requirements

Non-functional Requirements

Permissions

Dependencies

Acceptance Criteria

Edge Cases

Future Enhancements

Analytics Events

API Requirements

Database Tables

AI Behaviour

UI Components

Risks

Open Questions

Example Module

Authentication

Purpose

Allow users to securely access Atlas.

Functional Requirements

Students can register using:

- Email
- Google
- GitHub
- Apple ID

Students can:

- Login
 - Logout
 - Reset Password
 - Verify Email
 - Enable Two-Factor Authentication
-

Non-functional Requirements

Login response under 2 seconds.

99.9% uptime.

Encrypted credentials.

OAuth compliant.

User Story

As a student,

I want to login securely,

So that I can continue my learning progress.

Acceptance Criteria

✓ User receives verification email.

✓ OAuth login succeeds.

✓ Session restored.

✓ Invalid credentials handled.

✓ Account lock after repeated failures.

Analytics

Track

Registration

Login

Drop-off

Verification

Active Sessions

Future

Passkeys

SSO

Organization Login

7. Mission Engine

The Mission Engine is Atlas's core feature.

Responsibilities

- Deliver engineering missions
- Track progress
- Unlock next stages
- Manage Simplify Mode
- Trigger AI feedback
- Run hidden tests
- Award XP

Every mission belongs to one:

- Track
 - Module
 - Topic
 - Difficulty
 - Engineering Level
-

Mission Metadata

Mission ID

Mission Name

Difficulty

Track

Estimated Time

XP Reward

Prerequisites

Skills Tested

DSA Concepts

Framework Concepts

Industry Scenario

Mission Type

AI Prompt

Guide Steps

Hint Levels

Hidden Tests

Success Conditions

Portfolio Artifact

8. Mission Types

Logic Mission

Feature Mission

Architecture Mission

Bug Fix Mission

Optimization Mission

Refactoring Mission

Security Mission

Accessibility Mission

Testing Mission

Deployment Mission

System Design Mission

Interview Mission

9. Learning Tracks (Phase 1)

Programming Fundamentals

Java

Kotlin

Android

Jetpack Compose

Git

Databases

SQL

REST APIs

Future Tracks

Swift

iOS

Flutter

React

Next.js

Node.js

Spring Boot

Python

Go

Rust

DevOps

Cloud

AI/ML

Cybersecurity

Game Development

10. MVP Scope

Included

- Authentication
- Dashboard
- Learning Paths
- Lessons
- Mission Engine
- Simplify Mode
- Guide Mode
- Hint System
- Code Editor
- Compiler
- Hidden Tests
- XP
- Portfolio
- AI Review
- Admin Panel

Excluded

- Team Collaboration
 - Hiring Platform
 - Live Competitions
 - Marketplace
 - Voice Mentor
 - Mobile Applications
-

11. Risks

- Overly ambitious MVP
- High infrastructure cost
- AI hallucinations

- Challenge content quality
 - Scaling compiler infrastructure
 - Maintaining multiple language runtimes
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12. Success Metrics

Technical

- Crash-free sessions
- API latency
- Compiler success rate

Learning

- Lesson completion rate
- Mission completion rate
- Guide Mode usage
- Hint usage
- Knowledge retention

Business

- DAU
 - WAU
 - MAU
 - Premium conversion
 - Student retention
 - Portfolio completion
 - Interview success rate
-

Appendix A – Product Roadmap

Phase 1

Foundations



Phase 2

AI Learning



Phase 3

Engineering Simulator



Phase 4

Collaboration



Phase 5

Hiring Platform



Phase 6

Atlas Ecosystem